

GIT BASICS ASSIGNMENT 2

1. Github repository url: [Click here!](#)
2. Create a new repository on local in an empty directory (NOTE: do not clone)

```
[→ Desktop mkdir git-tasks
[→ Desktop cd git-tasks
[→ git-tasks ls -la
total 0
drwxr-xr-x  2 ankushhh28  staff   64  3 Oct 06:55 .
drwx-----@ 35 ankushhh28  staff 1120  3 Oct 06:55 ..
[→ git-tasks git init
Initialized empty Git repository in /Users/ankushhh28/Desktop/git-tasks/.git/
[→ git-tasks git:(main) ls -la
total 0
drwxr-xr-x  3 ankushhh28  staff   96  3 Oct 06:59 .
drwx-----@ 35 ankushhh28  staff 1120  3 Oct 06:55 ..
drwxr-xr-x  9 ankushhh28  staff  288  3 Oct 06:59 .git
[→ git-tasks git:(main) cd .git
[→ .git git:(main) ls -la
total 24
drwxr-xr-x  9 ankushhh28  staff  288  3 Oct 06:59 .
drwxr-xr-x  3 ankushhh28  staff   96  3 Oct 06:59 ..
-rw-r--r--  1 ankushhh28  staff  137  3 Oct 06:59 config
-rw-r--r--  1 ankushhh28  staff   73  3 Oct 06:59 description
-rw-r--r--  1 ankushhh28  staff   21  3 Oct 06:59 HEAD
drwxr-xr-x 16 ankushhh28  staff  512  3 Oct 06:59 hooks
drwxr-xr-x  3 ankushhh28  staff   96  3 Oct 06:59 info
drwxr-xr-x  4 ankushhh28  staff  128  3 Oct 06:59 objects
drwxr-xr-x  4 ankushhh28  staff  128  3 Oct 06:59 refs
→ .git git:(main) █
```

Ques: What changes did you observe in the current directory after initializing local repo?

=> A new .git directory is added to the local repository which contains all the things like config, description, hooks, HEAD etc.

- Point remote of your local git repo to the newly created repo on Github

```
[→ .git git:(main) git remote -v
[→ .git git:(main) git remote add origin git@github.com:ankushh2004/new-repo.git
[→ .git git:(main) git remote -v
origin  git@github.com:ankushh2004/new-repo.git (fetch)
origin  git@github.com:ankushh2004/new-repo.git (push)
```

- Print current git configs on terminal

```
[→ .git git:(main) git cogig --list
git: 'cogig' is not a git command. See 'git --help'.

The most similar command is
    config
[→ .git git:(main) git cofig --list
git: 'cofig' is not a git command. See 'git --help'.

The most similar command is
    config
[→ .git git:(main) git config --list
[→ .git git:(main) git config --local --list
→ .git git:(main)
```

```
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
remote.origin.url=git@github.com:ankushh2004/new-repo.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
(END)
```

- Update git config with your name and email address (only for current repository)

```
[→ .git git:(main) git congig --local user.name "Ankush's Assignment 2"
git: 'congig' is not a git command. See 'git --help'.

The most similar command is
    config
[→ .git git:(main) git config --local user.name "Ankush's Assignment 2"
[→ .git git:(main) git config --local user.email "asanodiya@ancilar.com"
[→ .git git:(main) git config --local --list
→ .git git:(main)
```

```
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
remote.origin.url=git@github.com:ankushh2004/new-repo.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
user.name=Ankush's Assignment 2
user.email=asanodiya@ancilar.com
(END)
```

- Create a file named "test1.txt"
- Added line with text "first line in file" in "test1.txt".
- Push this file to your Github repo.

```
[→ git-tasks git:(main) echo "first line in file" > test1.txt
[→ git-tasks git:(main) ✕ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test1.txt

nothing added to commit but untracked files present (use "git add" to track)
[→ git-tasks git:(main) ✕ git add .
[→ git-tasks git:(main) ✕ git commit -m"first line added"
[main (root-commit) 7dfb4b5] first line added
 1 file changed, 1 insertion(+)
 create mode 100644 test1.txt
[→ git-tasks git:(main) █
```

```
[→ git-tasks git:(main) git status
On branch main
nothing to commit, working tree clean
[→ git-tasks git:(main) git push origin main

Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 245 bytes | 245.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ankushh2004/new-repo.git
 * [new branch]      main -> main
[→ git-tasks git:(main)
[→ git-tasks git:(main) █
```

- Append line with text "second line in file" in "test1.txt".
- Add file to the staging area.
- Check the status of the file in all staging area
- Append line with text "Third line in file" in "test1.txt". (Note status of files across different areas).
- Remove "test1.txt" from the staging area.
- Create commit with message "My First Commit"
- Push commit to the remote repo

NOTE: checklist commits pushed to the remote

```
[→ git-tasks git:(main) echo "second line in file" >> test1.txt
[→ git-tasks git:(main) * cat test1.txt
first line in file
second line in file
[→ git-tasks git:(main) * git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test1.txt

no changes added to commit (use "git add" and/or "git commit -a")
[→ git-tasks git:(main) * git add .
[→ git-tasks git:(main) * git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   test1.txt

[→ git-tasks git:(main) * echo "third line in file" >> test1.txt
[→ git-tasks git:(main) * git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   test1.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test1.txt

[→ git-tasks git:(main) * git restore --staged test1.txt
[→ git-tasks git:(main) * git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test1.txt

no changes added to commit (use "git add" and/or "git commit -a")
[→ git-tasks git:(main) * git add .
[→ git-tasks git:(main) * git restore --staged
fatal: you must specify path(s) to restore
[→ git-tasks git:(main) * git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   test1.txt

[→ git-tasks git:(main) * git commit -m"my first commit"
[main 17c8834] my first commit
1 file changed, 2 insertions(+)
[→ git-tasks git:(main) git status
```



```

git-tasks git:(main) git status
On branch main
nothing to commit, working tree clean
git-tasks git:(main) git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ankushh2004/new-repo.git
 7dfb4b5..17c8834  main -> main
git-tasks git:(main) git log
git-tasks git:(main) git log --oneline
git-tasks git:(main)

```

```

17c8834 (HEAD -> main, origin/main) my first commit
7dfb4b5 first line added
(END)

```

- Update message of last commit "My First Commit" to "My Second Commit" (without creating a new commit)

```

git-tasks git:(main) git commit --amend -m"my second commit"

[main f3a4786] my second commit
Date: Fri Oct 3 07:23:19 2025 +0530
1 file changed, 2 insertions(+)
git-tasks git:(main) git log --oneline
git-tasks git:(main)

```

```

f3a4786 (HEAD -> main) my second commit
7dfb4b5 first line added
(END)

```

- Push commit with the updated message to the remote repo

```

→ git-tasks git:(main) ls
test1.txt
[→ git-tasks git:(main) git status
On branch main
nothing to commit, working tree clean
[→ git-tasks git:(main) git push origin main --force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 293 bytes | 293.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ankushh2004/new-repo.git
+ 17c8834...f3a4786 main -> main (forced update)
[→ git-tasks git:(main) git log --oneline
→ git-tasks git:(main) █

```

```

f3a4786 (HEAD -> main, origin/main) my second commit
7dfb4b5 first line added
(END) █

```

- Check the difference between the last and second last commits.

```

[→ git-tasks git:(main) git log --oneline
[→ git-tasks git:(main) git diff 7dfb4b5 f3a4786
[→ git-tasks git:(main) git diff HEAD~1 HEAD
→ git-tasks git:(main) █

```

```

diff --git a/test1.txt b/test1.txt
index f581c70..f3c05a1 100644
--- a/test1.txt
+++ b/test1.txt
@@ -1,3 @@
 first line in file
+second line in file
+third line in file
(END) █

```

- Revert the last commit and check the updated commit logs history.

```

git-tasks git:(main) git revert HEAD
[main 0a75c81] Revert "my second commit"
1 file changed, 2 deletions(-)
git-tasks git:(main) git log --oneline
git-tasks git:(main)

```

```

0a75c81 (HEAD -> main) Revert "my second commit"
f3a4786 (origin/main) my second commit
7dfb4b5 first line added
(END)

```

- Append the line with text "Fourth line in file" in "test1.txt" and commit it.
- Update the last commit such that the "test1.txt" also contains "Fifth line in file" (without creating a new commit)
- Push the updated commits to the remote repository

```

git-tasks git:(main) echo "fourth line in file" >> test1.txt
git-tasks git:(main) * cat test1.txt
cat: test1.txt: No such file or directory
git-tasks git:(main) * cat test1.txt
first line in file
fourth line in file
git-tasks git:(main) * git add test1.txt
git-tasks git:(main) * git commit -m"fourth line added"
[main aa0e664] fourth line added
1 file changed, 1 insertion(+)
git-tasks git:(main) git revert HEAD
[main 59edccd] Revert "test1.txt also contains fifth line in file"
1 file changed, 1 deletion(-)
git-tasks git:(main) git log --oneline -1
git-tasks git:(main) git log --oneline -1
git-tasks git:(main) git status
On branch main
nothing to commit, working tree clean
git-tasks git:(main) git puch origin main
git: 'puch' is not a git command. See 'git --help'.

The most similar command is
push
git-tasks git:(main) git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (7/7), 689 bytes | 689.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:ankushh2004/new-repo.git
f3a4786..59edccd main -> main
git-tasks git:(main)

```

```

59edccd (HEAD -> main) Revert "test1.txt also contains fifth line in file"
(END)

```

- Append some content in "test1.txt" file
- Pull all the content from remote repository to local repository
- Clone the remote repository to some other directory in your local system.

```
[→ git-tasks git:(main) echo "ading random line in file" >> test1.txt
[→ git-tasks git:(main) * cat test1.txt
first line in file
ading random line in file
[→ git-tasks git:(main) * git pull origin main
From github.com:ankushh2004/new-repo
 * branch          main          -> FETCH_HEAD
Already up to date.
[→ git-tasks git:(main) * ..
[→ Desktop mkdir clone-repo
[→ Desktop cd clone-repo
[→ clone-repo git init
Initialized empty Git repository in /Users/ankushhh28/Desktop/clone-repo/.git
[→ clone-repo git:(main) git clone git@github.com:ankushh2004/new-repo.git
Cloning into 'new-repo'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 11 (delta 2), reused 10 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), done.
Resolving deltas: 100% (2/2), done.
[→ clone-repo git:(main) * ls
new-repo
[→ clone-repo git:(main) * cd new-repo
[→ new-repo git:(main) ls
test1.txt
[→ new-repo git:(main) cat test1.txt
first line in file
[→ new-repo git:(main) █
```